

Polynomials – 8

Several variables and the polynomial method

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Theoretical minimum.

Theorem 1 (the grid lemma). If $P \in F[x_1, \dots, x_n]$ has degree at most d_i in x_i and vanishes on a grid $A_1 \times \dots \times A_n$ with $|A_i| > d_i$, then $P = 0$. In particular a non-zero polynomial over an infinite field is not the zero function, which fails over a finite field.

Theorem 2 (Combinatorial Nullstellensatz). Let $\deg P = d_1 + \dots + d_n$ and suppose the coefficient of $x_1^{d_1} \dots x_n^{d_n}$ in P is non-zero. Then for any sets A_i with $|A_i| > d_i$ there is a point of $A_1 \times \dots \times A_n$ at which P does not vanish.

Fact 1 (Schwartz–Zippel). A non-zero polynomial of total degree d over a field vanishes at no more than a $d/|A|$ fraction of the points of A^n for any finite A . This is the quantitative form of Theorem 1 and the reason random substitution tests polynomial identities.

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1. Prove the grid lemma, the Combinatorial Nullstellensatz and the Schwartz–Zippel bound, and show that each is sharp.
 2. (MIPT) A polynomial of degree d in n variables is given. Prove that either it takes a non-zero value at at least $2^n - d$ vertices of the cube $\{-1, 1\}^n$, or it vanishes at all of them.
 3. (Putnam 2003) Do there exist polynomials $a(x), b(x), c(y), d(y)$ such that $1 + xy + x^2y^2 = a(x)c(y) + b(x)d(y)$ holds identically?
 4. (IMC) Find all polynomials P in two variables with real coefficients satisfying $P(x, y)P(z, t) = P(xz - yt, xt + yz)$.

5. (MIPT) A polynomial $P(x, y)$ vanishes at exactly five points of the real plane. What is the smallest degree it can have?